STUDENT PERFORMANCE PREDICTION USING

MACHINE LEARNING

Abstract

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions without being explicitly programmed. In the education sector, machine learning can help analyze student data and predict academic performance .Predicting student performance helps teachers and educational institutions understand the factors that affect students' academic success. By analyzing factors such as study hours, attendance, and previous exam scores, it is possible to build a predictive model that estimates a student's final performance. This project focuses on predicting student performance using a machine learning model. The system analyzes input parameters related to student behavior and predicts the expected exam score. Python programming language is used to implement the system and the Scikit-learn library is used for machine learning algorithms. The model analyzes the relationship between study habits and exam scores. The results show that study hours and attendance significantly influence student performance. The system demonstrates how machine learning techniques can support data-driven decisions in the education sector

Introduction & Objectives

1. Introduction

Machine Learning is an important field of Artificial Intelligence that allows computers to learn from data and make predictions. It has applications in many fields such as healthcare, finance, business, and education. Educational institutions collect large amounts of student data including attendance, assignment scores, and exam results. Analyzing this data can help predict student performance and improve learning strategies. Student performance prediction helps teachers identify students who may need additional academic support. By using machine learning techniques, patterns in student behavior can be analyzed to estimate academic outcomes.This project focuses on building a predictive model that analyzes study hours, attendance, and previous exam scores to estimate a student's final exam performance.

2. Objectives of the Project

The main objectives of this project are:

- To analyze factors affecting student academic performance.
- To apply machine learning techniques for predicting exam scores.
- To build a prediction model using Linear Regression.
- To visualize the relationship between study habits and performance

Literature Review & Methodology

3. Literature Review

Educational data mining has become an important research area in recent years. Researchers use machine learning techniques to analyze student data and predict academic outcomes.Various algorithms such as Linear Regression, Decision Trees, Random Forest, and Support Vector Machines have been applied to

predict student performance.These models analyze factors such as study hours, attendance, assignment completion, and previous exam results. The analysis helps educators identify students who require additional academic support. Machine learning techniques allow educational institutions to make better decisions based on data analysis.

Methodology

The methodology used in this project follows a systematic machine learning workflow. The process includes several stages such as data collection, data preprocessing, feature selection, model training, and model testing. Each step plays an important role in building an effective prediction model for student performance.

4.1 Data Collection

Data collection is the first step in developing a machine learning model. In this project, a dataset containing student academic information is collected. The dataset includes important attributes related to student study behavior and performance.The dataset contains the following information:

- Study hours per day
- Attendance percentage
- Previous exam scores
- Assignment marks
- Final exam score

These attributes are selected because they have a direct influence on the academic performance of students. Each row in the dataset represents an individual student's academic record. The dataset provides the necessary information for training and evaluating the machine learning model.

4.2 Data Preprocessing

Data preprocessing is a crucial step in the machine learning process. Raw data collected from different sources may contain errors, missing values, or inconsistent formats. Therefore, the data must be cleaned and prepared before it can be used for model training.

The preprocessing steps include:

- Removing duplicate entries
- Handling missing values
- Converting data into proper numerical format
- Normalizing or scaling data if required

Cleaning the data improves the quality of the dataset and ensures that the machine learning algorithm can learn meaningful patterns from the data.

4.3 Feature Selection

Feature selection involves identifying the most important variables that influence the prediction outcome. In this project, several attributes such as study hours, attendance percentage, and previous exam scores are selected as input features.Selecting relevant features helps improve model performance and reduces

unnecessary complexity. By focusing on important variables, the machine learning model can make more accurate predictions.The selected features are used as input variables for training the prediction model.

4.4 Model Training

Model training is the process of teaching the machine learning algorithm to recognize patterns in the dataset. The dataset is divided into two parts:

- Training dataset
- Testing dataset

The training dataset is used to train the machine learning model. In this project, the **Linear Regression algorithm** is used to build the prediction model.Linear Regression is suitable for predicting continuous numerical values such as exam scores. The algorithm learns the relationship between input variables and the target variable by analyzing the training data.During the training phase, the algorithm calculates coefficients that represent the influence of each feature on the final prediction.

4.5 Model Testing

After training the model, it is important to evaluate its performance. This is done using the testing dataset, which contains data that was not used during the training phase.The trained model predicts the exam scores for the testing data. These predicted values are then compared with the actual exam scores to measure the accuracy of the model.Evaluation metrics such as prediction error and accuracy are used to determine how well the model performs. If the model produces accurate predictions, it indicates that the system has successfully learned the relationship between study habits and academic performance.This testing process ensures that the machine learning model is reliable and capable of predicting student performance effectively.

FlowDiagram

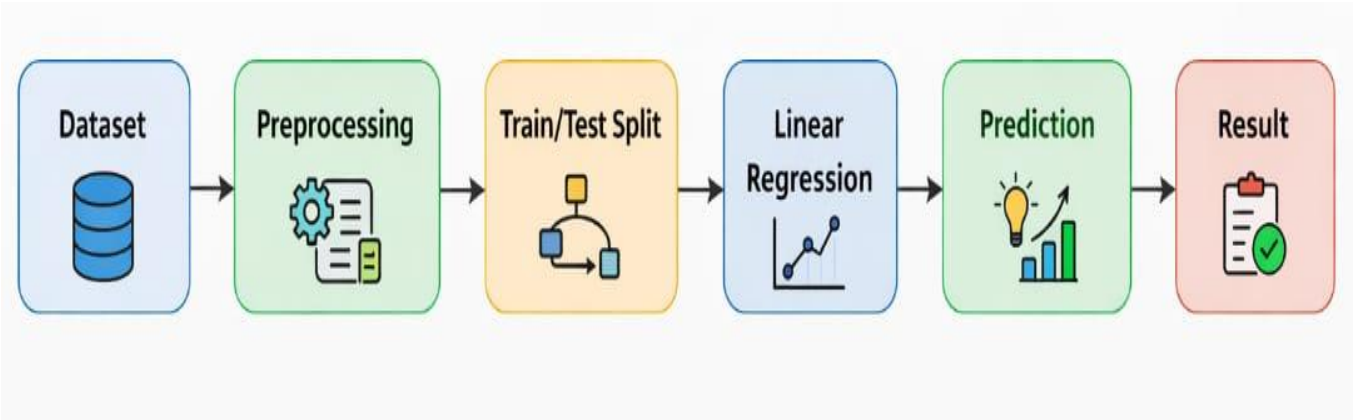


Diagram Explanation

1. Dataset

This is the first step in the project.In this stage, student data is collected**.**

Example data:

- ✓ Study hours

- ✓ Attendance percentage
- ✓ Previous exam marks
- ✓ Assignment marks

This dataset is used as the input for the machine learning model.

2. Preprocessing

In this step, the collected data is cleaned and prepared for training the model.

Tasks performed:

- ✓ Removing incorrect data
- ✓ Handling missing values
- ✓ Selecting important features

Data preprocessing helps to improve the accuracy of the model.

3. Train / Test Split

1. The dataset is divided into two parts:
2. Training data (80%) – used to train the model
3. Testing data (20%) – used to test the model performance
4. This helps to check how well the model predicts new data.

4. Linear Regression Model

In this stage, the Linear Regression algorithm is applied.

The model learns the relationship between:

- ✓ Input features (study hours, attendance, marks)
- ✓ Output (final exam score)

The model creates a mathematical relationship to predict results.

5. Prediction

After training the model, it is used to predict student performance.

Example: If a student studies 5 hours per day, the model predicts the expected exam score.

6. Result

This is the final stage.The system displays:

- ✓ Predicted marks
- ✓ Student performance

Tools, Implementation & Results

- **Tools and Technologies Used**

The following tools are used in this project:

- ✓ Programming Language
- ✓ Python

- **Libraries**

- ✓ Pandas – data analysis
- ✓ NumPy – numerical operations
- ✓ Matplotlib – data visualization
- ✓ Scikit-learn – machine learning algorithms

- **Development Environment**

- ✓ Jupyter Notebook

6. System Implementation

The dataset is first loaded using the Pandas library. Data preprocessing is performed to clean the dataset and remove inconsistencies. Next, the dataset is divided into training data and testing data. The Linear Regression algorithm is then applied to train the prediction model. The trained model learns the relationship between input variables such as study hours and exam scores. After training, the model can predict the final score of a student based on the input features.

7. Results and Discussion

After training the model, predictions are generated for the test dataset. The predicted scores are compared with actual exam scores.The results show that students who spend more time studying and maintain good attendance tend to achieve higher exam scores.Graphs can be used to visualize relationships between variables such as study hours and exam results.

Advantages, Limitations & Conclusion

8. Advantages of the System

- Predicts student academic performance
- Identifies factors influencing exam scores
- Helps teachers monitor student progress
- Supports data-driven decision making in education

Limitations

- Requires accurate and sufficient data
- Predictions depend on dataset quality
- Some personal factors affecting performance are not included in the model

CODE:

```
# Student Performance Prediction using Python

# Import libraries

import pandas as pd
```

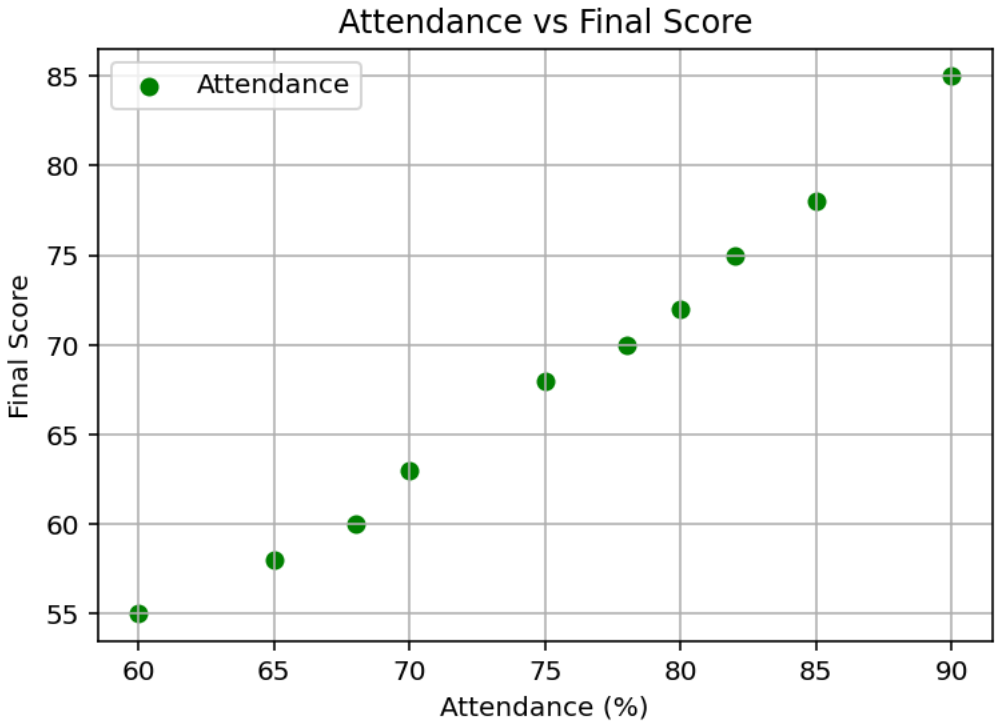
```
import numpy as np

import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_squared_error, r2_score

data = {

    'StudyHours': [2,3,4,5,6,7,8,3,5,6],

    'Attendance': [60,65,70,75,80,85,90,68,78,82],

    'PreviousScore': [50,55,60,65,70,75,80,58,68,72],

    'FinalScore': [55,58,63,68,72,78,85,60,70,75]

}

df = pd.DataFrame(data)

print("Dataset Preview:")

print(df)

X = df[['StudyHours', 'Attendance', 'PreviousScore']]  # Input features

y = df['FinalScore']                                   # Target variable

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42

)

model = LinearRegression()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("\nPredicted Scores:")

print(y_pred)

mse = mean_squared_error(y_test, y_pred)

r2 = r2_score(y_test, y_pred)

print("\nModel Evaluation:")

print("Mean Squared Error:", mse)

print("R2 Score:", r2)

plt.scatter(df['StudyHours'], df['FinalScore'], color='blue', label='Study Hours')
```

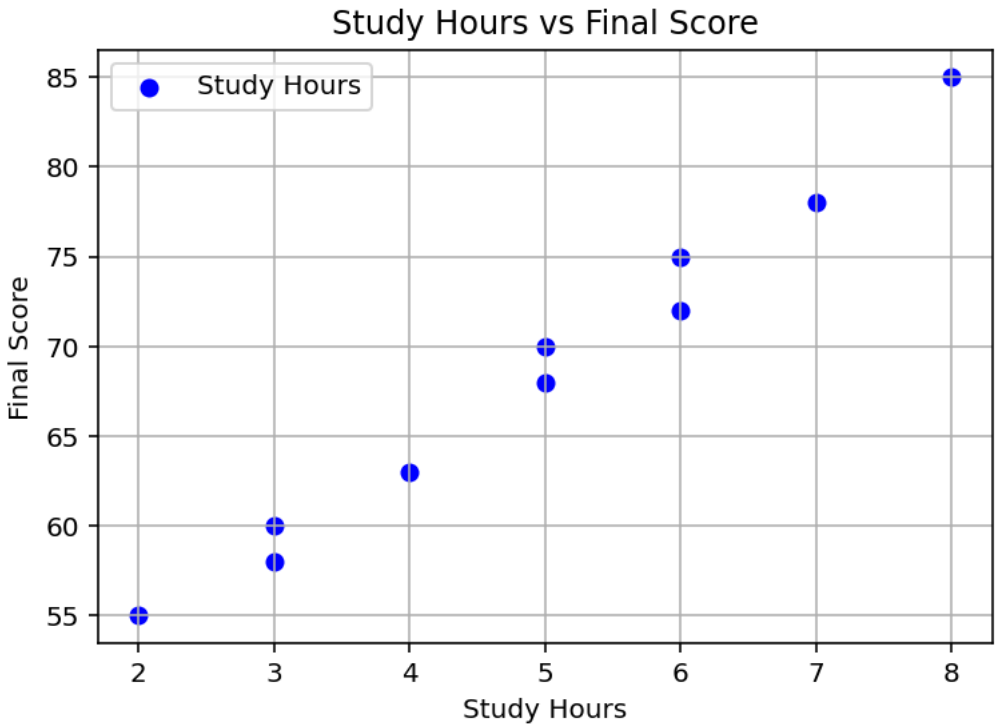
```
plt.xlabel("Study Hours")

plt.ylabel("Final Score")

plt.title("Study Hours vs Final Score")

plt.grid(True)

plt.legend()

plt.show()

plt.scatter(df['Attendance'], df['FinalScore'], color='green', label='Attendance')

plt.xlabel("Attendance (%)")

plt.ylabel("Final Score")

plt.title("Attendance vs Final Score")

plt.grid(True)

plt.legend()

plt.show()
```

OUT PUT

```
Dataset Preview:
   StudyHours  Attendance  PreviousScore  FinalScore
0           2          60             50          55
1           3          65             55          58
2           4          70             60          63
3           5          75             65          68
4           6          80             70          72
5           7          85             75          78
6           8          90             80          85
7           3          68             58          60
8           5          78             68          70
9           6          82             72          75

Predicted Scores:
[70.18199052 58.63127962]
```

```
Model Evaluation:
Mean Squared Error: 0.21581725477864147
R2 Score: 0.9940050762561489
```





Conclusion

This project presents a machine learning approach to predict student performance using a Linear Regression model. The system analyzes factors such as study hours, attendance, previous marks, and assignment marks to estimate the expected exam score. Data preprocessing and train–test splitting are used to improve model performance. The results show that machine learning techniques can help in predicting student academic performance effectively. This system can help teachers and students understand learning patterns and improve academic outcomes.